



Emotional Objects

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Contents

I	Introduction and Comparative Analysis	3
1	Introduction	3
2	Related Work	4
2.1	“The creation and development of an emotional robot inspired by Japanese art: R-Oiwa” (Dr. Giovanni Cappellari)	4
2.2	“Emotion-Driven Architecture for Human-Robot Interaction” (Dr. Eleonora Righi)	4
2.3	“Odile: an expressive robotic agent for emotional exchange and information sharing” (Dr. Erica Panelli)	4
3	Innovations Achieved by the Project	5
II	Project Development	6
4	Project Development Stages	6
4.1	Object Selection	6
4.2	Emotional Design	6
4.3	Feasibility Analysis	7
4.4	Structural and Functional Design	7
4.4.1	The Head	8
4.4.2	The Body	8
4.5	Requirements Analysis	8
4.6	Laboratory Access	9
4.7	Product Prototyping	9
4.8	Hardware Prototyping	9
4.9	Software development	10
4.10	Integration	10
4.11	Finalization	11
5	Tools and Technologies	11
5.1	Hardware	11
5.1.1	Microcontroller Selection	11
5.1.2	Sensors (Inputs)	11
5.1.3	Actuators (Outputs)	12
5.2	Software	12
5.3	Workshop Equipment	13
6	Challenges and Solutions	13
6.1	Non-Blocking Code Design	13
6.2	Coaxial Movement	14

III	Specification Documents	15
7	Prototyping Drafts	15
8	Operational Design	16
9	Emotional State Machine	18
10	Circuit Diagram	19
11	Software	20
11.1	Program structure	20
11.2	System Architecture	20
11.3	Core Execution Loop	21
IV	Results and Conclusions	22
12	Results	22
12.1	System Verification	22
12.2	User Experience	22
12.3	Gallery	24
13	Conclusions	26

Part I

Introduction and Comparative Analysis

1 Introduction

The “Emotional Objects” project explores the intersection between pragmatic product design and robotic art. The goal of this project is to take an everyday object and turn it into an “emotional” entity, capable of simulating vitality through expressive actions and situational reactions.

While modern technology often focuses on smart devices that can offer improved efficiency and utility to the user, this project aims to enrich the user experience in a different way. By imbuing an inanimate object with a distinct personality, a wider and richer range of interactions arises. Consequently, the use of said object is no longer limited to its bare utilitarian function. While the latter is not necessarily improved, the object is enhanced nonetheless, by granting it additional and innovative functionalities in areas such as education, therapy, or entertainment. The use of a common item creates a familiar setting for the user, which can be exploited in creative ways for the aforementioned purposes.

To achieve this new layer of possibilities, one must tap into the basics of affective computing and robotics. The former is needed to understand how human emotions can be simulated by machines in an effective way, while the latter is used to translate said emotions into mechanical operations.

The design, the implementation, and the results of the project will be presented in the following pages, first by outlining a general roadmap and later by focusing on the noteworthy details.

The development of the project commenced in February 2026 and concluded in June 2026. The research and fabrication were carried out within the Artificial Intelligence and Robotics Laboratory (AIRLab) at Politecnico di Milano, under the academic supervision of Professor Andrea Bonarini (Full Professor at the Department of Electronics, Information, and Bioengineering - DEIB).

2 Related Work

The realization of emotional robots is one of the many areas researched within the AIRLab, where the “Emotional Objects” project was developed. Three master’s theses, supervised by Professor Andrea Bonarini, have been selected to showcase previous work done in the field and various ways to approach the subject.

2.1 “The creation and development of an emotional robot inspired by Japanese art: R-Oiwa” (Dr. Giovanni Cappellari)

While the chosen object does not necessarily fall within the everyday item category, the goal behind *R-Oiwa* is almost entirely shared. The Japanese lantern is reimaged as a *Yokai*, which is a Japanese supernatural being, as seen in many artistic depictions. The project does not merely create a robotic art installation, but it also aims “*to discover how a ghostly lantern robot can express Ekman’s basic emotions while being constrained by an unusual shape and complex kinematics. Movements, lights, and voice lines were selected as the main features to try and achieve satisfying interactions with people.*” [1]

2.2 “Emotion-Driven Architecture for Human-Robot Interaction” (Dr. Eleonora Righi)

Despite not having an object as its starting point, the main purpose of the robot remains the same: displaying emotions, once again referring to Ekman’s core set. This robot is not intended for everyday use, instead, it can dynamically interact with a theatrical setting, in particular an improvisation performance. The project focuses on “*allowing the robot to adapt its responses based not only on the current stimulus but also on what has previously occurred. As a result, the interaction acquires temporal continuity and avoids purely stimulus-response dynamics, fostering more believable and context aware behavior.*” [3]

2.3 “Odile: an expressive robotic agent for emotional exchange and information sharing” (Dr. Erica Panelli)

The last work mentioned explores the difficulty of conveying expressions in a non-anthropomorphic robot, a problem which the “Emotional Objects” project also faces. As stated by the author: “*We set ourselves several constraints during the execution of this project. To ensure that the focus was on movement and non-verbal communication, we chose a robot form that was not human-like. We also did not want to use any other type of feedback, visual or sound.*” [2] While some of these constraints do not apply to our case, the challenge is still shared. *Odile* is not autonomous but controlled, yet its core functionality is the ability to communicate, which directly correlates to showing an emotion.

3 Innovations Achieved by the Project

As the previous examples demonstrate, affective computing and emotion-driven robotics boast an extensive amount of literature and research. Therefore, the innovation pursued by the project does not lie in developing new hardware but in making creative and unique use of already available technologies.

The core concept is mimicking human emotions, but from a completely different point of view. Personal experience and psychology make our sentiments sufficiently easy to define, but applying the same feelings to a non-humanoid and normally inanimate entity requires a particular thought process.

The challenge of the project is not posed by extreme mechanical complexities, but by the subtle nuances of human perception. The effort shifts toward imagining an immersive and dynamic experience, and making it as effective and as understandable as possible for the average user. In a comparison with the aforementioned projects, the question remains the same: how can a robot convey emotion? Nonetheless, the answer is drastically different. A lantern and a uniquely designed robot vary incredibly in their display of the same state. In our project, where the chosen object is a grandfather clock, the same problem applies.

For this reason, the very core of the project demands a form of innovation. Of course, there are some overlaps with the other projects. For instance, the technological components used in *R-Oiwa* [1] are similar, but their use can vary. As psychology suggests, the color blue is a universal indicator of sadness; therefore, both projects exploit this subconscious clue to symbolize this emotion, yet after this initial similarity the projects diverge. A lantern and a grandfather clock do not share any structural similarity. The manipulation of their parts for expressive movements demands different approaches.

There is also another aspect of the “Emotional Objects” project that is unique to its goal and even more so to its chosen object. While other projects focus on creating a robot with artistic value or with a specific function (e.g., communication), the realization of an emotional everyday object requires the preservation of its basic utility (in our case, of course, the telling of the time). This calls for an analysis tailored to the chosen item, which needs to blend into a domestic environment while maintaining its personality. There is also the implicit limit of the object’s shape, which is widely recognized and therefore cannot be excessively altered to fit possible mechanical needs.

In conclusion, the transformation of the grandfather clock into an emotional entity offers unique opportunities and constraints; thus, it represents an innovation in itself. As a matter of fact, we have not found a comparable example of an already existing item with the same characteristics.

Part II

Project Development

4 Project Development Stages

4.1 Object Selection

The first step needed to approach a project such as this was narrowing down its scope and refining the general idea into a polished and usable design. The number of everyday objects to choose from is nearly infinite; therefore, one must define a series of criteria to guide the selection. Although imagination is the one true limit, there are many other aspects that demand consideration.

First of all, the chosen object must be big enough to accommodate at least the microcontroller (which has a significant volume). This excludes the majority of common utensils (e.g., cutlery, stationery, etc.) that would require custom-made technologies that surpass the scope of this project. The chosen object should also be easily buildable or alterable over many iterations, since the project calls for a fair amount of prototyping. We can now exclude large pieces of furniture or appliances that would slow down the process. The last criterion to evaluate is the technology of the chosen object. Since the goal of the project is to improve an existing object without completely impairing its original function, it would be sensible to avoid complex items. For instance, transforming a coffee maker, while not impossible, would require a difficult integration with the existing software and hardware, with little to no room for error, given the costly nature of the item that would not be easily replaceable.

Having considerably trimmed the list of eligible items, we can now move onto less functional requirements. The peculiar nature of our objective demands a creative analysis of the selection. To streamline this second filtering, we can use an additional criterion: the chosen object must present a good amount of movable parts to favor expressive dynamism in the final product.

With all these required properties in mind, we whittled down the initial broad idea and selected a viable object: a grandfather clock. It has sufficient internal room for many components, it can be crafted, and its main function, which is telling the time, requires only rudimentary technology. It also checks the final quality: it has an abundance of moving parts, like the hands, the pendulum, and the many rotating disks that can be found on its dial.

4.2 Emotional Design

Once the object had been decided, the next step was defining its emotional properties. In order to achieve a coherent behavioral design, a personality was required. Thus, we ideated the following character: an enigmatic and odd grandfather clock that can be amused just as easily as it can be irritated. It seeks attention and is characterized by extreme reactions. Despite not disliking the presence of the user, it tolerates it only for short periods of time. It also makes it somewhat difficult to understand its needs, but it can reward those who decipher them. Essentially, the feeling that we wanted to convey to the user is that of interacting with a ghostly entity who is reluctant to cooperate, yet not necessarily evil.

This led to the distinction between an idle state, where the goal of the object is to attract passersby, and an active state that triggers when a user is interacting with the object, in which

the emotions can be displayed. During the idle state, the design choices are focused on making the clock as weird and eye-catching as possible. This has been achieved in subtle ways that convey a general feeling of oddness: the hands spin counterclockwise, the minute hand ticks every second, the hour hand ticks every minute, and every once in a while the hands start spinning wildly and the pendulum swings faster. A user whose attention has been caught will then experience one of the following emotions:

- **Anger:** A noisy state that can be exited only by appeasing the clock's needs.
- **Sadness:** A slow-moving state that can be exited by playing a game with the clock.
- **Hunger:** A mysterious state where the clock tries to scare the user.
- **Happiness:** A hard-to-obtain state where the clock becomes more compliant with the user's interactions.

These high-level ideas guided a more detailed design that defined possible features the clock could implement and their placement in the object.

4.3 Feasibility Analysis

Due to the creative nature of the previous stage, some of the ideas could not be fully translated into a working model. The emotional design was therefore subjected to a thorough feasibility analysis which revealed aspects that needed to be scrapped, and others that needed further refinement. This overview took into consideration the following criteria:

- **Microcontroller and Memory Capacity:** Microcontrollers, although powerful, still present hardware limits that need to be taken into consideration. The extent of the unfiltered design did not prove to be excessive for the computing and storing capacities of most microcontrollers.
- **Power Management:** Every component added to the circuit increases the overall power consumption of the system. To ensure the project remains energy-efficient and compatible with standard, readily available power supplies, the total number of components was optimized and reduced.
- **Mechanical and Structural Constraints:** Limiting the dimensions of the product to a tabletop version of a grandfather clock greatly affected the space available. Thus, some non-essential features had to be removed in order to free space for the main components.
- **Authenticity and User Experience:** The weirdness of the emotional clock should not overtake its normal counterpart. This meant that some features, although interesting, had to be reimaged or removed because they could not be immediately understood by the user or easily disguised. Furthermore, some features required elements that would not be normally found on a clock and were scrapped to keep the object as authentic as possible.

4.4 Structural and Functional Design

With the analysis completed, the core features were defined. The finalized design of the object presents two sections. To see a visual representation of the following elements, please refer to the [Prototyping Drafts](#) section.

4.4.1 The Head

It represents the core of the object's emotionality. It takes the dial of the clock and turns it into an expressive face. It exploits all the elements normally present on it:

- The charging holes house LEDs to represent glowing eyes.
- The hands can either spin (always counterclockwise, like previously stated, to increase the odd factor) or stop at set positions to simulate expressions, like furrowed eyebrows or a mustache.
- The brand, usually written on the dial, is placed on a rotating disk as a curved inscription, which can appear either as a smile or a pout. This offers a perfect opportunity to add another detail to the object. The fictional brand "*Tempus Mordet*" (which translates to "*Time Bites*") is wordplay on the famous Latin quote "*Tempus Fugit*" ("*Time Flies*").
- The lunar dial can rotate to show different images associated with various emotions: a sunny sky for happiness, stormy clouds for anger, and a starry night for calm or sadness.

4.4.2 The Body

It grants more immersive user interactions. Its appearance is reminiscent of a gaping jaw with a lolling tongue. It is made up of two main parts:

- A base equipped with an opening compartment, which the clock will autonomously open to encourage interaction during specific emotional states. It contains colored buttons that unlock different reactions based on the current emotional states. The sad state offers a game that consists of memorizing a color sequence of increasing length and repeating it by pressing the buttons of the corresponding color. The colors are shown by the LED eyes. Meanwhile, in the happy state, pressing the yellow button will cause the hands to spin wildly, pressing the red button will cause the pendulum to swing fast, pressing the green button will cause the eyes to glow in a rainbow cycle, and pressing the blue button will cause the clock to show the current time. Leaving the compartment open will force the angry state during the next interaction.
- A highly mobile pendulum. In addition to normal oscillations at various speeds, it is capable of performing a curving motion as if reaching towards the user. This ability is used during the hungry state to surprise the user.

4.5 Requirements Analysis

By combining the emotional, structural, and functional designs, it became possible to define a precise list of requirements that had to be considered during the ensuing stages.

- The system must autonomously monitor its immediate perimeter environment to allow user detection.
- The hands need to mimic the continuous rotation of normal clock hands and also stop at target positions.
- The system must transition seamlessly between states based on multiple factors.

- The system must handle multiple concurrent movements at once.
- The pendulum must be made with flexible material to allow the desired curvature without losing firmness during normal oscillations.
- The system must detect the state of its opening compartment, discerning between closed and open, and also be able to autonomously open it.
- The system must accurately track time even when not in use to show it during user interaction.

4.6 Laboratory Access

Given the hardware-heavy nature of the project, once the designing phase was finalized, further development required proper equipment. It became therefore fundamental to operate in a specialized infrastructure such as the Artificial Intelligence and Robotics Laboratory (AIR-Lab) at Politecnico di Milano. Access to the AIRLab requires the reading of documents listing [safety norms](#) and [rules](#), and the completion of the following mandatory online safety courses: “*Corso Base sulla Sicurezza - 'Formazione generale'*”, “*Formazione specifica per lavoratori - basso rischio*”, “*Corso sulla sicurezza nei Laboratori di Robotica del DEIB*”.

4.7 Product Prototyping

After a final revision of the project design with the supervisor Prof. Andrea Bonarini, the necessary materials and components were gathered to begin the prototyping phase, which relied heavily on the workshop equipment of the laboratory. In order to check the theorized proportions, a prototype of the casing was produced. To allow easy alterations and reduce production costs, the material selected for this purpose was cardboard.

The object was crafted from scratch, with a reference model that can be found in the [Operational Design](#). The fairly complex structure of the grandfather clock demanded multiple adjustments, and the precise needs of some mechanisms (e.g., the clock face) required the use of metal rods. Lastly, the pendulum was realized using a metal spring with a nylon string tied to its end that can control its curvature.

4.8 Hardware Prototyping

The hardware prototyping followed an incremental process, using a solderless breadboard to simplify adjustments. First, every component was tested individually to spot faulty ones and verify proper wiring. Then the architecture was divided into smaller and more easily testable units. A detailed list of the components used can be found in the [Hardware](#) section, while the following paragraphs use general mentions of the elements.

- **User Detection:** To test this, the emotional states were replaced by an LED glowing with different colors, and a stub program was written to test continuous proximity and absence detection. This unit was also used to experiment with various data sanitation methods, required when working with sensors prone to noise errors like the ultrasonic kind.
- **Expressions:** The clock face comprises multiple motors that are orchestrated in configurations needed to form the emotional expressions. By isolating the clock face, it was

possible to easily calibrate the single motors with the right angles and speeds. This unit required precise placements for its parts, which were achieved thanks to a custom-made stand crafted with cardboard. The independent coaxial movement of the hands (see [Coaxial Movement](#)) was also successfully tested in this unit.

- **Pendulum:** Despite being one of the smallest units component-wise, its correct implementation required multiple iterations. After crafting a custom stand for the motors, careful calibration was needed to obtain realistic oscillatory movement (which is represented by a sine wave).
- **User Input:** The four buttons hidden in the opening compartment were first tested on a solderless breadboard alongside a single LED to simulate the basic implementation of the minigame played during the sad emotional state. To guarantee structural integrity during repeated user interaction and reduce the required space, the buttons were later soldered to a perfboard, inserted into a corresponding slot of the casing.

4.9 Software development

The code written during the hardware prototyping phase was fragmented and test-specific. Having established the requirements of the individual units, the next step was to define the overarching architecture. Thus, custom data structures were defined (e.g., the struct tracking servo motor data), along with finite state machines and highly reconfigurable constants needed to streamline the integration process.

Other than improving the readability of the code with enums and variable names, the focus of this stage was implementing the concurrent movements mentioned in the requirements, that had been momentarily ignored in the testing of the hardware prototyping in favor of a more direct procedural approach (more on this matter in [Non-Blocking Code Design](#)).

4.10 Integration

With the firmware fully programmed to regulate the clock's behavior, the focus shifted to the structural integration and assembly of the hardware components. The casing, still in cardboard, was assembled with high-duty adhesive to withstand heavier loads, and the custom stands were firmly integrated into its structure. While ensuring firm integrity, particular attention was given to maintenance access. Where possible, magnets were used instead of adhesive so that the casing could be easily taken apart and reassembled if one needed to modify its internal circuit.

After securing each component in its correct spot, the wiring proved to be the main challenge of this stage. The limited available space caused inevitable overlaps that were mitigated by a distinctive use of the colors of the jumper wires.

Running the software on the full system revealed bugs and unwanted behaviors that had to be fixed either by rewriting parts of the code or reconfiguring the constants defined for this purpose. Increased power consumption called for optimizations in the component management, for instance, by dynamically attaching and detaching servo motors in order to eliminate idle current draw and minimize unpredictable jittery behavior. While the non-blocking design proved to be effective, the general architecture needed some fine-tuning, for example, by refining conflicting state machine configurations.

4.11 Finalization

Once the integration phase reached an overall satisfactory result, despite the hardware limitations, the project marched towards completion with the aesthetic overhaul of the final product. The bare casing was covered using colored cardboard and wood grain vinyl sheets, acquiring a realistic and vintage look meant to enhance the user experience. For the same purpose, the pendulum was wrapped in a fabric layer. The completed project was presented to the supervisor with a demonstration of its features.

5 Tools and Technologies

5.1 Hardware

The following paragraphs will illustrate hardware choices and their motivation. For a more detailed overview of their implementation within the system, please refer to the [Circuit Diagram](#).

5.1.1 Microcontroller Selection

To accommodate the hardware requirements of the project, a comparative analysis of standard prototyping microcontrollers was conducted. The criteria that guided the choice ranged from the number of available digital/PWM pins to operating voltage compatibility with standard hobby components. Table 1 outlines the technical specifications of the considered boards against the requirements of the project.

Table 1: Comparative Analysis of Microcontroller Boards

Specification	Arduino Uno	ESP32	Arduino Mega 2560	Requirement
Digital I/O Pins	14	36	54	9
PWM Channels	6	16	15	10
Flash Memory	32 KB	4 MB	256 KB	Medium (≈ 16 KB)
Operating Voltage	5V	3.3V	5V	5V Preferred

It appears evident that the Arduino Uno cannot physically meet the project requirements due to its low PWM pin count. While the ESP32 offers superior memory and processing speeds, its 3.3V operating voltage requires logic-level shifters to control standard 5V servos and sensors, unnecessarily increasing circuit complexity. Therefore, the Arduino Mega 2560 was selected as the optimal choice.

5.1.2 Sensors (Inputs)

- **Ultrasonic Distance Sensor (HC-SR04):** This component grants the ability to detect an approaching user by measuring the return time of echo signals sent at fixed intervals. Despite the 400 cm range capacity, its use within the project was limited to a proximity sensor, which would not be affected by hardware limitations.
- **Digital Pushbuttons:** These components were adopted to register user input and facilitate interaction. The possibility to add colored caps and the recognizable button shape simplify the user interface.

- **Microswitches:** While employing a similar technology to the pushbuttons, the slight mechanical differences of the microswitches (mainly, the minimal pressure needed for their activation and their miniature size) are better suited for machine-to-machine contact. These components were used to detect specific states of the system, like the position of the compartment, or to act as physical triggers needed to home the clock hands into desired positions.
- **Hall Effect Sensor (A3144):** This sensor detects the presence and strength of a magnetic field using the Hall effect. In this project, it served as a non-contact alternative to mechanical microswitches for hand homing. For a detailed reasoning of the use of two different sensor types for the same purpose, please refer to the [Coaxial Movement](#) section.
- **Real-Time Clock (RTC) Module (DS3231):** This component is an external chip connected to the Arduino board via its integrated I^2C interface (i.e., using only the specialized Serial Data (SDA) and Serial Clock (SCL) lines). It provides accurate time and, thanks to its external battery, can precisely track time also when disconnected or when the system is not powered.

5.1.3 Actuators (Outputs)

- **Positional Micro Servos (SG90):** The majority of the moving parts were realized using this type of motor. The servos provide precise angular control ($0^\circ - 180^\circ$) that can be utilized to orchestrate the many configurations of elements such as the lunar dial, the brand disk, the pendulum, and the string pulling mechanism. A micro servo paired with a tiny magnet was also employed to create a locking mechanism for the opening compartment: by tucking the magnet away, the compartment is released and its contents are exposed.
- **Continuous Rotation Micro Servos (FS90R):** The remaining moving parts (i.e., the hands) present another requirement that cannot be met by standard micro servos. Clock hands boast seamless 360° rotation, which was achieved by introducing this special type into the assets. Despite the name, continuous rotation servos' behavior is closer to that of a DC motor since they do not offer positional control and can instead only be commanded in speed and direction. This produced the need for the aforementioned homing mechanism provided by the microswitch and Hall effect sensor.
- **Common-Cathode RGB LEDs:** Despite having more demanding pin needs compared to their monochromatic alternative, RGB LEDs provide a full spectrum of colors through independent PWM control of the three additive primary color channels (Red, Green, Blue). This enables a more vibrant emotional characterization of the clock's eyes.

5.2 Software

The software was coded within the Arduino IDE, a developing environment specialized in compiling and uploading code to Arduino hardware. The intuitive user interface and real-time serial communication tools simplified the debugging process by allowing the microcontroller to display internal states in a readable format.

The project also employed the following external libraries to improve management of the used peripherals:

- **Servo.h:** Manages the generation of pulse-width modulation (PWM) signals for positional servos, allowing precise degree-based control.
- **NewPing.h:** Replaces the standard blocking ultrasonic sensor communication methods with optimized, non-blocking echo timing routines, ensuring smoother execution of concurrent movements.
- **Wire.h:** Manages the I^2C serial communication protocol used by the RTC module.
- **RTCLib.h:** Facilitates data retrieval from the external RTC module with methods such as `now()` that return a nicely formatted date and timestamp.

The final firmware is contained in a `.ino` file, the standard extension for Arduino source code. The program follows the procedural paradigms of the C programming language to maintain low-level control over the system logic. However, the external libraries written in C++ required the occasional use of object-oriented syntax, such as dot notation, needed to invoke functions and instantiate objects defined by the drivers.

5.3 Workshop Equipment

The fabrication of the physical prototype necessitated extensive utilization of the AIRLab workshop facilities. Beyond standard crafting utensils (e.g., box cutters, pliers, screwdrivers, hand saws, files, etc.), the assembly process required specialized equipment, specifically a drill press and a soldering station, whose use warranted detailed explanation.

6 Challenges and Solutions

The project faced numerous challenges during its development that required the redesign of software or structural elements. Some have already been illustrated in previous sections, like the dynamic attachment and detachment of servo motors to optimize power consumption and the homing logic applied to non-positional motors. Further explanations will be provided here for the remaining challenges.

6.1 Non-Blocking Code Design

The test code written during the [Hardware Prototyping](#) stage made extensive and naive use of the `delay(ms)` function to control the components. While sequential updates could be accepted in a test version, the final design required tasks to be executed concurrently. The blocking call of the `delay(ms)` function completely froze the program execution, guaranteeing reception of the signals to the actuators at the price of blinding the microcontroller to inputs received by the sensors.

This was overcome by introducing custom software timers evaluated against the non-blocking `millis()` function, which returns the total number of milliseconds that have elapsed since the microcontroller began executing the current program. With this global timestamp, the firmware assumed an asynchronous scheduling architecture.

For this to work, the program stores the timestamp of each task's last update and fixed intervals of execution in dedicated variables. During each iteration of the main `loop()`, the system continuously evaluates the delta time using the following condition:

$$\Delta t = \text{currentMillis} - \text{lastUpdate} \geq \text{interval} \quad (1)$$

If this condition is met, the task is executed and its timer variable is updated to the current time; otherwise, the task is ignored. This non-blocking paradigm reduced latency and kept the microcontroller receptive to asynchronous external events signaled by the sensors.

To ensure long-term system integrity, all timing variables were declared as `unsigned long` data types. The unsigned 32-bit hardware counter that the `millis()` function polls overflows and resets to zero after ≈ 49.7 days of continuous runtime. By employing the subtraction paradigm ($\Delta t = \text{currentMillis} - \text{previousMillis}$), the firmware can rely on the arithmetic properties of hardware-level Two's Complement subtraction. Consequently, even when `millis()` returns an overflowed value, the condition is still correctly evaluated.

Moreover, to avoid dangerous power spikes, the servos were encapsulated in a custom struct, which stores the target angle and current angle in addition to the last update and execution interval. This permitted a more intelligent power management: the servos are updated with unitary increases of their angular positions instead of being forced to perform an ample and power-hungry sweep.

6.2 Coaxial Movement

One of the most recognizable elements of a traditional clock dial lies in the concentric rotation of its hands. Replicating this coaxial movement requires two independent actuators to rotate visual components along the exact same central axis. To achieve this, a custom pulley-driven transmission system was engineered.

Because the two servo motors cannot be physically stacked on top of one another, only one servo was kept directly aligned with the central rotation axis, while the other was dislocated diagonally. The rotational torque of the dislocated servo was transferred to the central axis using two identically sized pulleys (1:1 ratio) linked by an elastic rubber band. The minute hand was then fixed directly to the central servo via a screw, which simultaneously serves as the pivot for the primary pulley. The hour hand was mounted to this pulley with a metal extension, allowing synchronous rotation with the secondary pulley fixed to the dislocated servo.

While the 1:1 ratio between the pulleys should theoretically guarantee synchronized movement, empirical observation revealed that the rubber band introduces slight delays and staggers in the transmission, which over time accumulate into an unpredictable decoupling between the servo horn and the corresponding clock hand. If the central servo can rely on the synchronized position of the horn and the hand to use a microswitch, the homing mechanism of the dislocated servo (i.e., the hour hand) must instead be triggered by the hand itself. The fragility of the hand, which is also completely exposed to the user, required a hidden, non-contact trigger. Therefore, a tiny magnet was mounted to the hand, and a Hall effect sensor was positioned on the back of the clock face, activated by the passage of said magnet without mechanically obstructing the hand's movement.

Part III

Specification Documents

7 Prototyping Drafts



Figure 1: Neutral expression on the left and angry expression on the right.



Figure 2: Pendulum curvature idea demonstration.

8 Operational Design

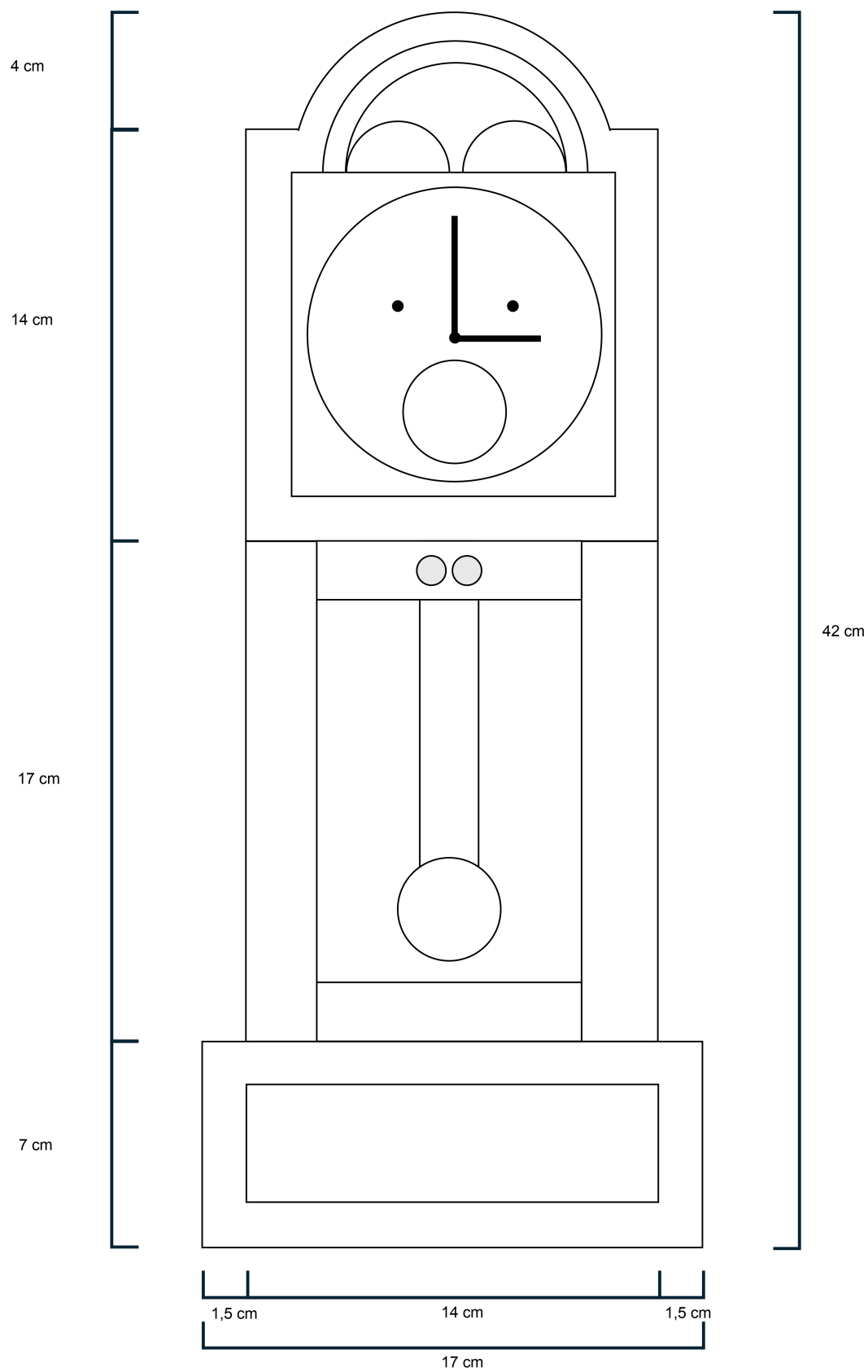
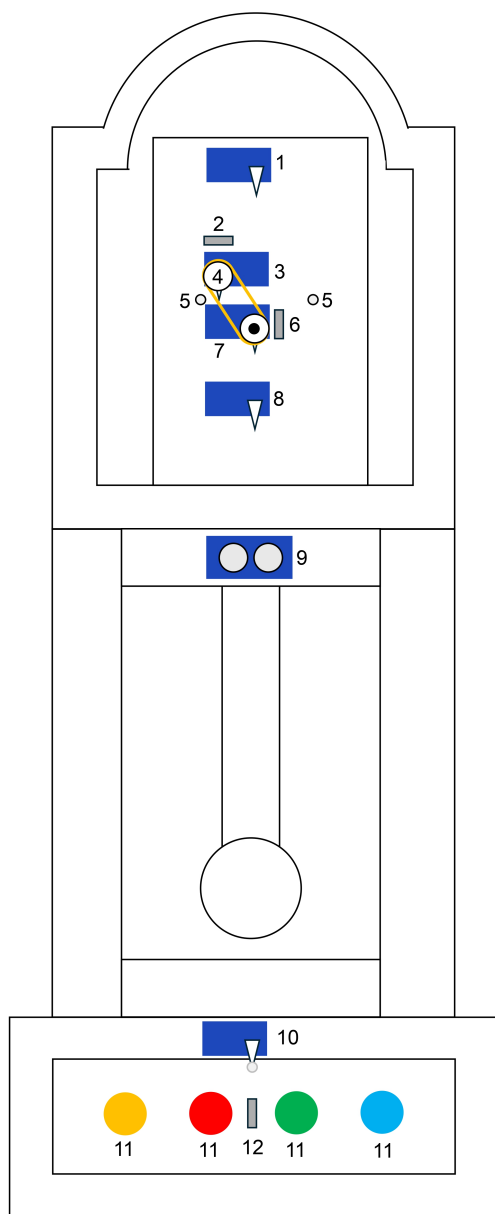
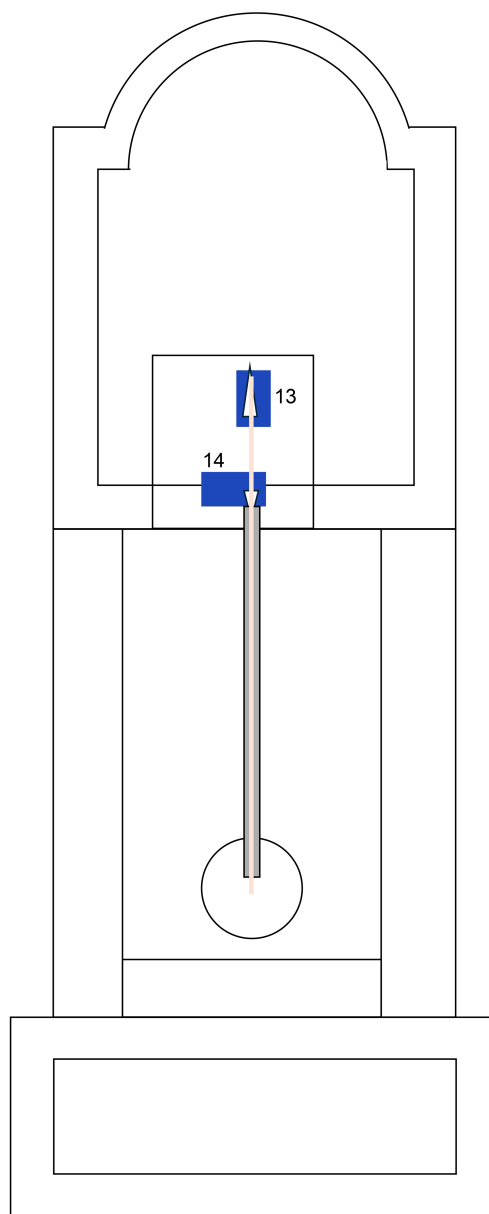


Figure 3: Full object model with dimensions of main elements.



(a) First layer.



(b) Second layer.

#	Component Type	Function
1	Micro Servo	Rotates the lunar dial
2	Hall Effect Sensor	Calibrates the hour hand for homing logic
3	Continuous rotation micro servo	Spins the hour hand via pulley
4	Pulley with rubber band	Transfers servo rotation to hour hand
5	RGB LED	Changes eye color based on emotional state
6	Microswitch	Calibrates the minute hand for homing logic
7	Continuous Rotation Micro Servo	Spins the minute hand
8	Micro Servo	Rotates the brand disk
9	Ultrasonic Sensor	Detects approaching users
10	Micro Servo with magnet	Opens and locks compartment
11	Pushbutton	Detects user input
12	Microswitch	Detects open and closed compartment state
13	Micro Servo	Moves pendulum
14	Micro Servo with nylon string	Pulls and releases string to curve pendulum

Figure 4: Internal positional schema with numbered components.

9 Emotional State Machine

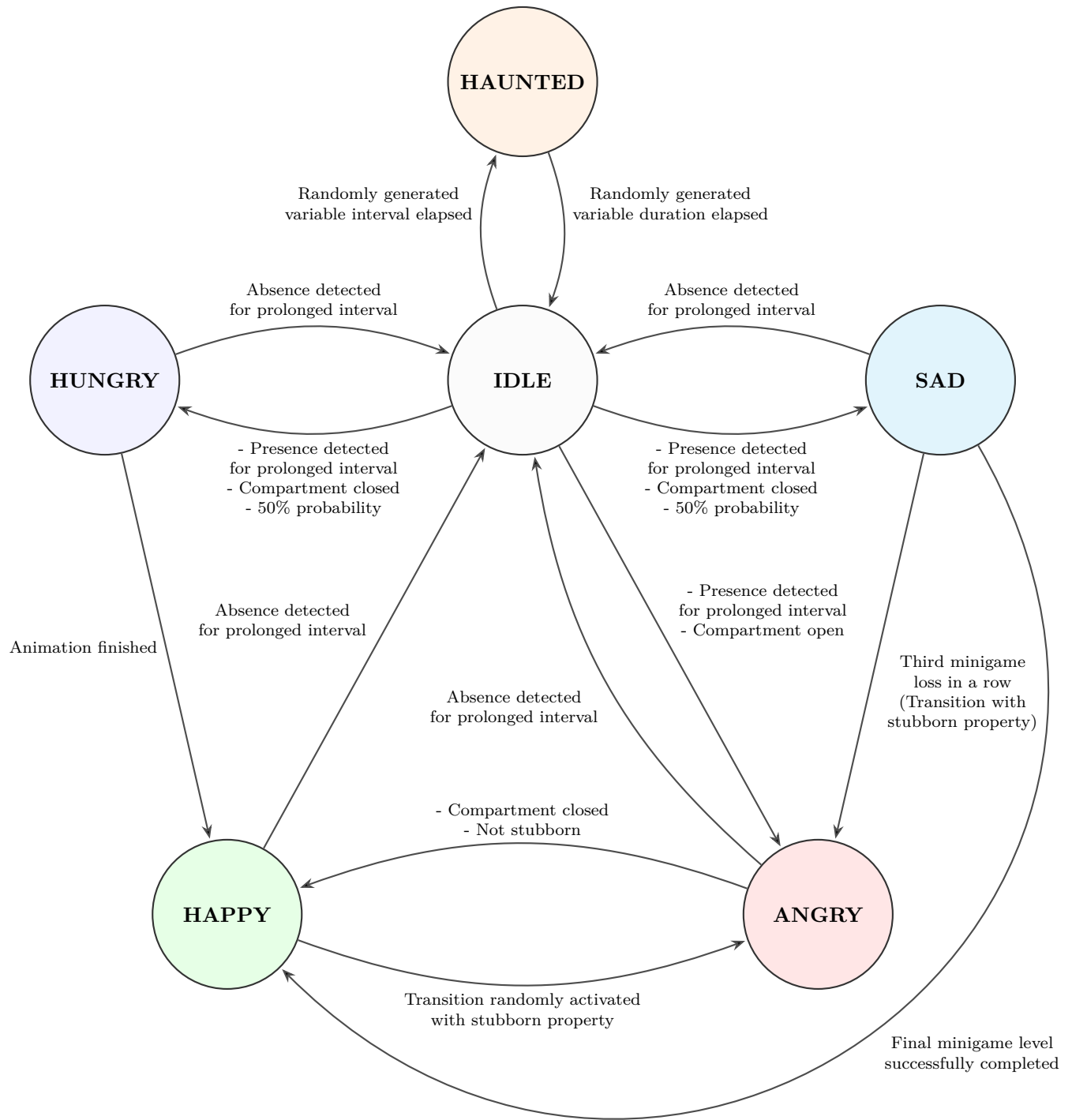


Figure 5: High-level Finite State Machine (FSM) representation of the clock's emotional behavior based on autonomous transitions and user interaction input.

10 Circuit Diagram

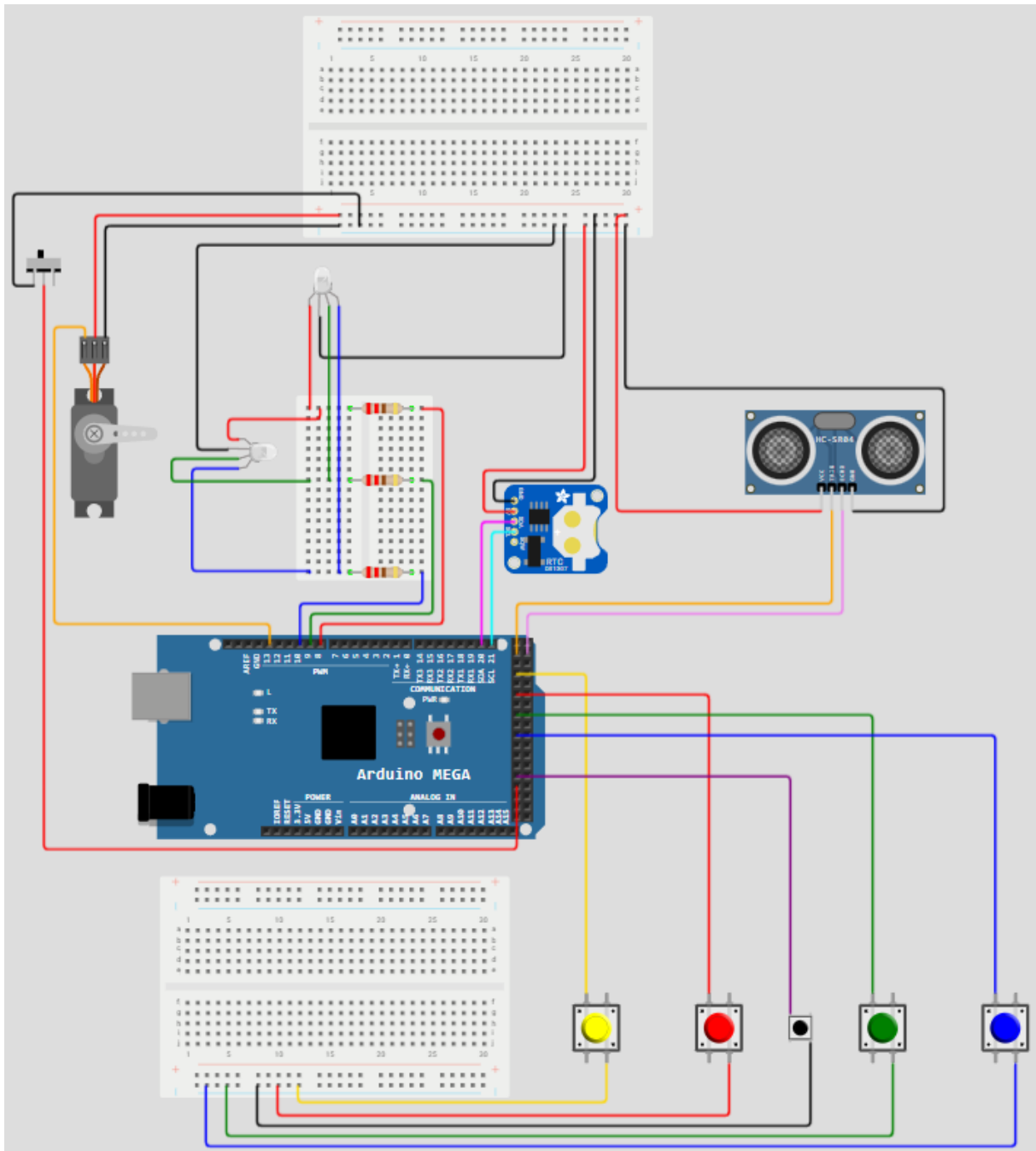


Figure 6: Simplified Circuit Diagram.

Figure 6 represents a high-level abstraction of the complete circuit scheme meant to showcase component integration with the central microcontroller. To favor legibility, most duplicate modules were removed to focus entirely on the wiring requirements of each peripheral type. Despite conceptual simplifications, core architectural elements have been preserved:

- **Multiple breadboards:** The actual circuit contains two separate power bus strips and a mini solderless breadboard used by the LEDs, as shown in the diagram. Multiple power rails were required by the spatially distinct sections of the object (the head and the base),

while the additional breadboard's purpose is purely related to cable management.

- **LED wiring:** As the diagram illustrates, the two RGB LEDs share the same PWM pins of the microcontroller. This choice halves the number of $220\ \Omega$ resistors and PWM channels needed by the circuit and ensures the desired synchronization of the eyes.

Although omitted from the diagram for simplicity, it is critical to highlight that the architecture utilizes two isolated power supplies to segment the electronic loads. The Arduino Mega 2560 is powered by a dedicated supply to ensure logic processing stability and prevent resets. Concurrently, an external AC-to-DC adapter delivers a 5V-3A supply to the power rails utilized by the entire system.

11 Software

The complete source code, accompanied by an operational manual, is hosted in the project's official GitHub repository, which can be accessed at the following URL:

<https://github.com/michelcostantini/Emotional-Objects-Tempus-Mordet>

11.1 Program structure

The firmware is divided into six sections:

1. **Pin Configuration:** Numerical pins are never directly referenced, but they are all wrapped inside constants to simplify future reconfigurations.
2. **Enumerations:** States and indexes are mapped to explicative names to improve readability. This includes servo names, button names and color codes.
3. **Configurable constants:** Similar to pins, many numerical properties of the system (such as timing intervals, speeds, or angular positions) are wrapped inside easily alterable constants. This approach minimizes the presence of “magic numbers”, which are notoriously hard to understand and update.
4. **Variables Declaration:** In addition to high-level states, every component stores specific data that details the actual state of the system and its last update timestamp, as illustrated in [Non-Blocking Code Design](#). Naturally, for every hardware peripheral, a software counterpart is declared.
5. **Helper Functions:** These functions offer a layer of abstraction between hardware-level signal control and software commands or information requests.
6. **State Machine Handlers:** While the core [Emotional State Machine](#) is regulated within the standard `loop()` function, the logic of the independent component sub-states is separated into dedicated handlers.

11.2 System Architecture

The global architecture is structured like a layered finite state machine. The concurrent nature of the updates allows the definition of specialized state machines also for independent units.

- **Core FSM:** Cycles between `NORMAL_MODE`, `HAUNTED_MODE`, and `EMOTIONAL_MODE` based on randomized events or user detection.
- **Emotional States:** Sub-states within the `EMOTIONAL_MODE` with different behaviors. States and transitions are outlined in the [Emotional State Machine](#) section.
- **Interactive Memory Game Machine:** A nested FSM executing during the sad emotion that handles sequence generation and validation logic (`GAME_GENERATE`, `GAME_FEEDBACK_OFF`), user input (`GAME_INIT`, `GAME_WAIT_INPUT`, `GAME_FEEDBACK_ON`), and visual animations (`GAME_AMUSED`, `GAME_SHOW_PLAY`, `GAME_SHOW_OFF`, `GAME_WIN_FLASH`, `GAME_LOSE_FLASH`).
- **Mechanical Hand Controller:** A hardware-level FSM designed to govern dynamic hand positioning. It comprises four main states:
 - `HAND_IDLE`, which is segmented into `TICK_IDLE` and `TICK_PULSING` sub-states to simulate a realistic incremental tick-and-stop movement.
 - `HAND_HOMING`, during which the servo triggers the corresponding sensor two times in order to self-calibrate the time required for a full 360° rotation, thereby mitigating errors induced by voltage fluctuations.
 - `HAND_OFFSETTING`, where the hand keeps spinning for a specified fraction of the complete rotation duration to achieve target configurations.
 - `HAND_READY`, which holds the actuator in position while waiting for user input.

11.3 Core Execution Loop

Within the `loop()` function that the microcontroller continuously calls, the execution is segmented into the following procedures, designed with the concurrent principles of [Non-Blocking Code Design](#):

- **Input Processing and Filtering Layers:** Updates the global system timestamp. It filters user input by awaiting stabilization of signals arriving from the pushbuttons over a debounce interval. Concurrently, raw inputs from the ultrasonic distance sensor are sanitized via a non-blocking Exponential Moving Average (EMA) filter to detect human presence.
- **User Permanence and Behavioral Trigger Handling:** Evaluates user presence or absence using physical and temporal thresholds. It also evaluates possible state transition triggers and specific conditions needed to define sub-states.
- **The Core State Machine:** Executes the behavioral branch associated with the current state by setting global variables during initial transitions or by dispatching dedicated handlers.
- **Actuator Control:** Orchestrates emotional kinematics by referencing the global variables updated in the previous steps. This procedure invokes the hand-specialized FSM, implements the locking and opening logic of the compartment, synthesizes harmonic pendulum oscillations by adopting a trigonometric sine wave, updates concurrent servo positions with staggered incremental steps, and detaches inactive motor pins which do not require torque to optimize power consumption.

Part IV

Results and Conclusions

12 Results

12.1 System Verification

The system's fulfillment of the requirements was verified by observing a full execution sequence upon booting the microcontroller. The prototype followed the intended behavior by correctly transitioning between states and operating concurrent tasks without resets. The full range of desired features was achieved:

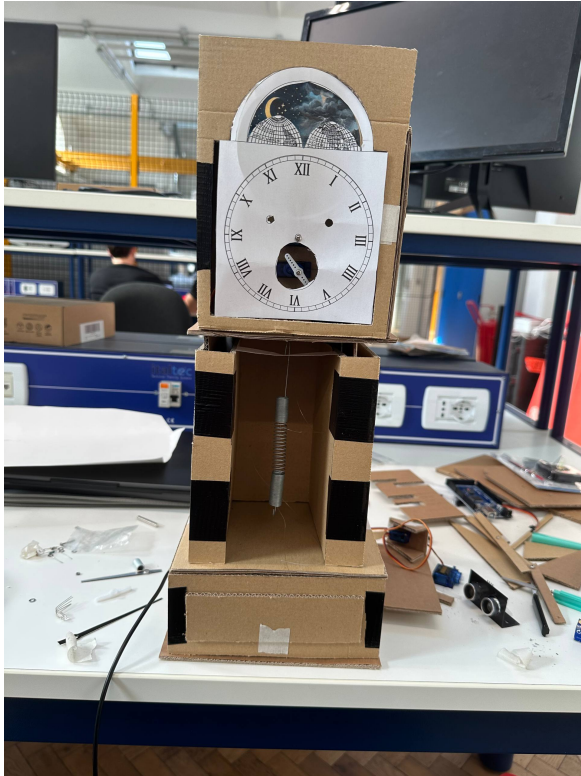
- **Initialization:** Upon power-up, the firmware correctly initiates the I^2C communication protocol with the real-time clock (RTC). If the chip is missing or unresponsive, the execution is halted as intended to prevent undefined responses during time polling.
- **Asynchronous Actuator and Sensor Tasks:** The main `loop()` function correctly schedules concurrent tasks, achieving smooth updates that can be observed in all the behavioral states. For instance, during `NORMAL_MODE` the minute hand ticks at one-second intervals while the pendulum follows its oscillating trajectory. Moreover, this does not blind the microcontroller from intercepting user detection signals fired by the ultrasonic sensor.
- **Proximity Detection:** The prototype successfully recognizes the presence and the absence of a user and complies with the configured thresholds.
- **Emotional Behavior:** The object effectively reacts to user interactions and randomized events. Transitions between emotional states are triggered by the intended conditions, like the state of the compartment or the mini-game result.
- **Expressiveness:** The clock displays a satisfactory range of expressions, which match the emotional state. With varying degrees of precision, the clock face can assume every desired configuration. The torque-demanding state of the taut string (corresponding to the curved pendulum) is correctly exempted from the detaching logic applied to other components.
- **Mini-Game Execution:** Under the sad emotion, the memory mini-game promptly reacts to user input. The execution transitions smoothly between animation and game phases, triggering emotional responses to wins and losses.
- **Happy State User Controls:** The happy emotion effectively fulfills the commands requested by the user using the colored pushbuttons, including the time-telling function synchronized with the RTC.

12.2 User Experience

After completing final test runs, the prototype was ready to be exposed to users who were not aware of its functionalities. The interaction with new users highlighted the following aspects of the design:

- **Oddity and Curiosity:** Users immediately noted the counterclockwise movement of the clock hands, which instantly drew their attention. What they initially assumed to be a mechanical or assembly error proved, instead, to be an intentional, eye-catching feature.
- **Replayability and State Variation:** Given the design choices guiding the behavioral patterns, different test users experienced varying interaction timelines. However, because the randomness of the sad and hungry states was unmitigated, certain emotions occasionally repeated in immediate sequence. While this randomness was intended to reduce predictability, users unaware of the unseen states were less prone to continue interacting after experiencing a pattern repetition.
- **User Interface:** As expected, the interaction prompts generated by the clock proved somewhat difficult for uninformed users to decode. The “needs” of the clock are intentionally obtuse, requiring a degree of trial and error before they can be satisfied. While this personality trait enhances the poltergeist-like qualities of the object, it also introduced complexity. Nonetheless, when interacting with such a strange object, discovering the logic behind the unusual should be considered a key factor.
- **Reception of Displayed Emotions:** The emotional expressions of the clock were effectively conveyed to the users through visual clues. The diverse range of expressions was positively received, and features unexpected in a traditional timepiece, such as the pendulum curving to reach toward the user, registered as compelling surprises.

12.3 Gallery



(a) Initial Prototype.



(b) Final Product.

Figure 7: Comparative showcase of the product in two stages of development.



(a) Hungry State.



(b) Happy State with exposed buttons.

Figure 8: Full object showcase.



(a) Happy State.



(b) Angry State.



(c) Sad State.



(d) Hungry State detail: curved pendulum.

Figure 9: Emotional expressions showcase.

13 Conclusions

The development of the “Emotional Objects” project proved to be a stimulating and complex process, which encouraged the acquisition of numerous new practical skills and theoretical notions. The project boasted a mixture of software, hardware, and design elements that culminated in a tangible and operating final product. Combining these distinct disciplines required a meticulous balancing of theoretical models, mechanical constraints, and creative choices.

The manufactured prototype offered a satisfactory implementation of the project objective, meeting functional requirements during empirical testing and gathering positive user evaluations. Thanks to its emotion-driven design, the object developed a perceived personality that elevates it over its common counterpart.

Starting from this working baseline, future iterations could introduce even more features, expanding the expressiveness of the object. For instance, custom voice lines and reactive sound effects could be introduced by adding dedicated speaker modules (e.g., an external MP3 playback shield). Unlocking another sensory channel would surely create a more immersive experience that would improve conveyed feelings.

Alternatively, new emotional states could be implemented with entirely different environmental triggers. One example would be a “fear” emotion, activated when a user produces a sudden, loud noise. This could be achieved by integrating an acoustic sensor module.

During the project, we have witnessed the metamorphosis of an inanimate object into an entity capable of displaying nuanced emotions and real-time situational reactions. This result attests to the foundational principles of affective computing and immersive human-computer interaction. While the concept might appear inherently paradoxical, proper design can successfully grant machines the ability to simulate human-like emotions, even in non-anthropomorphic forms.

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